

TOBY S. COX

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EDUCATION

CLEMSON UNIVERSITY

Clemson, SC

Bachelor of Science in Computer Science, Minor in Artificial Intelligence

May 2027

Bachelor of Science in Mathematical Sciences, Minor in Physics

GPA: 4.00/4.00

RELEVANT COURSEWORK

Data Structures and Algorithms (A)	Theory of Probability (A)	Discrete Mathematical Structures (A)
Introduction to Software Engineering (A)	Artificial Intelligence (A)	Calculus of Several Variables (A)
Introduction to Ordinary Differential Equations (A)	Linear Programming (A)	Applied Data Science (A)
Machine Learning: Implementation and Evaluation (A)	Linear Algebra (A)	Linear Analysis (IP)

EXPERIENCE

Undergraduate Teaching Assistant

Clemson University

CPSC 1020 (Computer Science II)

Fall 2024

- Leading twice-weekly lab sections with thirty students, covering topics such as Linux, C++ syntax, data structures and algorithms, file input/output, classes, and inheritance
- Holding office hours to provide individualized assistance to students
- Grading student submissions for a class of over one hundred students

SELECTED PROJECTS

Hardware-Aware Quantum Circuit Synthesis (*Undergraduate Researcher, REU*)

Summer 2025

- Adapted diffusion models to generate hardware-aware quantum circuits
- Achieved upwards of 8x improvement in circuit fidelity over current methods
- Accepted to ACM/IEEE Supercomputing Conference 2025

FABRIC Operations- Accelerating Innovation and Research (*Undergraduate Researcher, REU*)

Summer 2025

- Leveraged the FABRIC testbed for national, high-scale scientific computing
- Analyzed data transfer rates using iPerf3 across different congestion control protocols

Hands-On Quantum Computing (*Undergraduate Researcher, Creative Inquiry*)

Spring 2025 – Present

- Conducted formal verification of quantum algorithms using Isabelle/HOL
- Implemented quantum algorithms in qBraid, using Qiskit library for Python

HackMIT (*Team Lead, Hackathon*)

Fall 2025

- Leveraged YOLO and MediaPipe to create a real-time synthesizer from paper
- Awarded 1st place in Suno's "Best Musical Hack" category, out of ~300 total teams at the event

HoyaHacks (*Team Lead, Hackathon*)

Spring 2026

- Created monitoring software for HPC clusters that tracks most accessed files, with interactive dashboard
- Awarded "Best Use of MongoDB" out of ~120 total teams at the event

iQuHack (*Participant, Hackathon*)

Spring 2025

- Implemented QUBO optimization on D-Wave's quantum computers
- Investigated speed-up of quantum computers over classical computers for this optimization task

NASA SpaceApps (*Team Lead, Hackathon*)

Fall 2025

- Modeled differential equations via approximation and iteration to model asteroid impacts and their effects
- Awarded 1st place in the 3rd biggest local event in the US for "Most Inspirational Hack"

SKILLS

PROGRAMMING LANGUAGES

- Python, C++, JavaScript, C, Java, R, Julia, Isabelle/HOL, HTML, CSS, TypeScript, Sage, SQL, Prolog

SOFTWARE/PACKAGES

- Google Suite, Microsoft Office, AWS, Postman, Qiskit, TensorFlow, PyTorch, NodeJS, MongoDB

MISCELLANEOUS

- German, LaTeX/Overleaf, Unix/Linux, qBraid

HONORS AND AWARDS

HackMIT 2025 "Best Musical Hack" (1st Place)

President's List (Clemson) (Fall 2023 – Present)

Chase Family Scholarship

Clemson University Honors College (Fall 2023 – Present)